CODE DOCUMENTATION

A Fuzzy Logic Restoration Scheme for Martian Imagery from Perseverance Rover

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Abstract

Martian photographs (MPs) captured by the Perseverance rover are exclusively used for studying the geology of Mars. Analysis of MPs is often difficult as the visibility of objects in certain regions of the MPs are masked by low light and shadow. Already available methods for improving the quality of MPs — namely, demosaicing, correction for dust, super-resolution, pansharpening, histogram equalization (HE), and its variants — do not directly address the poor visibility of objects caused by low light and shadow. To address this problem, a Fuzzy Transformation Function (FTF) is introduced to improve the visibility of MPs. The FTF models intensities in red, green, and blue colour channels of MPs as fuzzy sets, compresses the fuzzy membership values (FMV) at the extrema and stretches the intermediate values applying a nonlinear operator. Processed MPs are retrieved from the modified FMVs via defuzzification. On 100 MPs, FTF exhibited the highest Visual Saliency-Induced Index (VSI) (0.9861 ± 0.0026) and Sparse Feature Fidelity (SFF) (0.9801 ± 0.0055) and the lowest Lightness Order Measure (LOM) (117.8 ± 15.4605) and Lightness-Order-Error (LOE) (0.9 ± 1.4681) compared to the latest image enhancement techniques, implying that FTF improves the visibility of contents in MPs masked by low light and shadow without causing any information loss and colour distortions.

1. Overview

This document provides complete technical documentation for the MATLAB implementation of the Fuzzy Colour Image (FCI) Enhancement algorithm, also referred to as the Fuzzy Transformation Function (FTF) in the associated manuscript. The algorithm is designed to improve the visibility of low-light and shadow-masked regions in Martian photographs captured by NASA's Perseverance rover.

2. Motivation and Problem Statement

Martian photographs serve as the primary resource for geologic analysis of Mars. However, many images suffer from significant visibility degradation due to the following factors:

- Low-light conditions caused by the greater distance of Mars from the Sun
- Shadow masking arising from the rugged Martian terrain topology
- Poor object visibility in specific spatial regions of interest

Existing enhancement techniques including demosaicing, dust correction, super-resolution, pansharpening, histogram equalisation (HE) and its variants do not directly address these visibility issues in the fuzzy domain. The proposed FTF fills this gap by operating directly on fuzzy membership values across all three RGB colour channels.

3. Function Reference

3.1 FCI_Colour — Colour Image Wrapper

FCI_Colour is the top-level entry point. It applies the FCI single-channel enhancement to each of the three colour planes (R, G, B) of an RGB image independently and returns the enhanced colour image.

Signature

```
Enhanced_Image_Colour = FCI_Colour(Input_Image_Colour,
Exponential_Fuzzification_Parameter)
```

Input Parameters

Parameter	Description
Input_Image_Colour	uint8 array of size H×W×3. The input RGB Martian photograph to be enhanced.
Exponential_Fuzzification_Parameter	Scalar double (EFP). Controls the degree of fuzzification and the strength of contrast enhancement. A higher value produces stronger enhancement.

Output

Parameter	Description
Enhanced_Image_Colour	uint8 array of size H×W×3. The FCI-enhanced RGB image.

Internal Constant

Denominational_Fuzzification_Parameter = 127 (hardcoded). This is set to the midpoint of the 8-bit intensity range [0, 255] to provide a normalised fuzzification denominator.

3.2 FCI — Core Single-Channel Enhancement

FCI is the core processing function that implements the full three-stage fuzzy pipeline: fuzzification, intensification of fuzzy membership function, and defuzzification. It operates on a single image channel (H×W matrix).

Signature

```
Enhanced_Image = FCI(Input_Image, EFP, DFP)
```

Input Parameters

Parameter	Description
Input_Image	uint8 H×W array. A single colour channel extracted from the RGB image.
EFP	Exponential Fuzzification Parameter (scalar double). Shared with the parent call from FCI_Colour.
DFP	Denominational Fuzzification Parameter (scalar double). Passed as 127 from FCI_Colour.

Output

Parameter	Description
Enhanced_Image	uint8 H×W array. The enhanced single-channel image.

4. Algorithm Description

The FCI algorithm processes each pixel intensity through three sequential and mathematically invertible stages. Each colour channel is treated as an independent 2D matrix.

Stage 1 — Fuzzification

Each pixel intensity value x is mapped to a fuzzy membership value μ ∈ [0, 1] using the following transformation:

```
μ(x) = ( 1 + (X_max − x) / DFP ) ^ (−1 × EFP)
```

Where X_max is the maximum intensity in the current channel. Higher intensities (brighter pixels) yield membership values closer to 1; lower intensities (darker pixels) yield values closer to 0. The parameter EFP controls the steepness of this mapping.

Stage 2 — Intensification of Fuzzy Membership Function (IFMF)

A piecewise nonlinear operator is applied to compress membership values near 0 and 1 (extrema) while stretching intermediate values. This operation enhances contrast in mid-tone regions, which correspond to partially visible objects in shadow or low light:

```
IFMF(i,j) = 2 × μ(i,j)²                        if μ(i,j) ≤ 0.5
IFMF(i,j) = 1 − 2 × (1 − μ(i,j))²            if μ(i,j) > 0.5
```

This is a classical fuzzy intensification (INT) operator that redistributes membership values, emphasising the differentiation of intermediate fuzzy regions.

Stage 3 — Defuzzification

The enhanced pixel intensities are recovered from the modified fuzzy membership values by inverting the fuzzification transformation:

```
x' = X_max - ( ( (1 / IFMF)^(1/EFP) - 1 ) × DFP )
```

The result is clipped and cast back to uint8 (0–255), yielding the enhanced pixel intensities in the original domain.

5. Parameters

Parameter	Symbol	Default / Range	Description
Exponential Fuzzification Parameter	EFP	2 (recommended)	Controls the steepness of the membership function and the strength of the intensification. Higher values yield more aggressive enhancement. Typical range: 1–5.
Denominational Fuzzification Parameter	DFP	127 (fixed)	Normalises the pixel intensity range. Set to the midpoint of [0, 255] to ensure symmetrical fuzzification behaviour.

6. Usage Example

6.1 Basic Usage

```
% Read a Martian image
img = imread('martian_image.jpg');

% Apply FCI enhancement with EFP = 2
EFP = 2;
enhanced_img = FCI_Colour(img, EFP);

% Display original and enhanced images
figure;
subplot(1,2,1); imshow(img);            title('Original');
subplot(1,2,2); imshow(enhanced_img);   title('FCI Enhanced');

% Save output
imwrite(enhanced_img, 'enhanced_martian.jpg');
```

6.2 Batch Processing

```
files = dir('images/*.jpg');
for k = 1:length(files)
    img = imread(fullfile('images', files(k).name));
    out = FCI_Colour(img, 2);
    imwrite(out, fullfile('enhanced', files(k).name));
end
```

7. Requirements

Requirement	Details
MATLAB Version	R2016b or later. No additional toolboxes are required.
Input Image Format	Any format supported by MATLAB imread: JPEG, PNG, TIFF, BMP, etc.
Image Channels	Input must be a 3-channel RGB image for FCI_Colour. For single-channel processing, call FCI directly.
Data Type	Input is internally converted to double for computation and output as uint8.

8. Performance Metrics

The FTF was evaluated on 100 Martian photographs sourced from the Perseverance rover mission using four objective quality metrics:

Metric	Full Name	Desired Direction	FTF Score
VSI	Visual Saliency-Induced Index	Higher ↑	0.9861 ± 0.0026
SFF	Sparse Feature Fidelity	Higher ↑	0.9801 ± 0.0055
LOM	Lightness Order Measure	Lower ↓	117.8 ± 15.4605
LOE	Lightness Order Error	Lower ↓	0.9 ± 1.4681

The FTF achieved the highest VSI and SFF values and the lowest LOM and LOE values compared to all competing enhancement methods evaluated in the study. This indicates that FTF enhances the visibility of contents masked by low light and shadow without introducing information loss or colour distortions.

9. Citation

If you use this code or algorithm in your research, please cite the associated manuscript:

```
@article{shamsudeen_fuzzy_martian,
  title   = {A Fuzzy Logic Restoration Scheme for Martian Imagery
             from Perseverance Rover},
  author  = {Shamsudeen, Fousia M and Venugopal, Vipin and B N, Anoop},
  journal = {[Journal Name]},
  year    = {[Year]},
  volume  = {[Volume]},
  pages   = {[Pages]},
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}
```

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